

Ricardo Yarza

Publications

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For the most recent list, see any of the following: [NASA ADS](#), [Google Scholar](#), [ORCID](#). Underline indicates mentored student.

In press or under review:

4. Kotten, B., Soares-Furtado, M., **Yarza, R.**, Nine, A., et al.
Lithium Enrichment in a Subgiant Star with a Brown Dwarf Companion: A Planetary Engulfment Candidate
Submitted to AAS journals
3. Twum, A. A. G., Vigna-Gómez, A., MacLeod, M., Everson, R. W., et al. incl. **Yarza, R.** 2026, [arXiv:2602.10211](#)
The Landscape of Unstable Mass Transfer in Interacting Binaries and Its Imprint on the Population of Luminous Red Novae
2. Lane, K. T., Stephan, A. P., Soares-Furtado, M., Stassun, K. G., & **Yarza, R.** 2026, [arXiv:2601.00949](#)
Observable Metal Pollution in Main-Sequence Stars: Simulations of Rocky Planets Engulfed by Stars in the 0.5 to 1.4 $M_{\odot}$ Range
1. Soares-Furtado, M., Limbach, M. A., Vanderburg, A., Bally, J., et al. incl. **Yarza, R.** 2024, [arXiv:2406.01492](#)
The TEMPO Survey II: Science Cases Leveraged from a Proposed 30-Day Time Domain Survey of the Orion Nebula with the Nancy Grace Roman Space Telescope

Published:

14. **Yarza, R.**, MacLeod, M., Idini, B., Murray-Clay, R., & Ramirez-Ruiz, E. 2025, [ApJL](#), **995**, L48
Evolution of the ZTF SLRN-2020 Star-Planet Merger
13. Bush, R. C., Wu, S. C., Everson, R. W., **Yarza, R.**, et al. 2025, [ApJL](#), **990**, L7
Black Hole Survival Guide: Searching for Stars in the Galactic Center that Endure Partial Tidal Disruption
12. Liu, C., **Yarza, R.**, & Ramirez-Ruiz, E. 2025, [ApJ](#), **979**, 40
Repeating Partial Tidal Encounters of Sun-like Stars Leading to Their Complete Disruption
11. Hutchinson-Smith, T., Everson, R. W., Twum, A. A., Batta, A., et al. incl. **Yarza, R.** 2024, [ApJ](#), **977**, 196
Rethinking Thorne–Żytkow Object Formation: The Fate of X-Ray Binary LMC X-4 and Implications for Ultra-long Gamma-Ray Bursts
10. Rosselli-Calderon, A., **Yarza, R.**, Murguía-Berthier, A., Rohoza, V., et al. 2024, [ApJ](#), **977**, 16
The Evolution of Binaries Embedded Within Common Envelopes

9. Farias, D., Gall, C., Narayan, G., Rest, S., et al. incl. **Yarza, R.** 2024, [ApJ, 977, 152](#)
SN 2021foa: The “Flip-flop” Type II_n/I_{bn} Supernova
8. Ransome, C. L., Villar, V. A., Tartaglia, A., Gonzalez, S. J., et al. incl. **Yarza, R.** 2024, [ApJ, 965, 93](#)
SN 2023ixf in Messier 101: The Twilight Years of the Progenitor as Seen by Pan-STARRS
7. Capistrant, B. K., Soares-Furtado, M., Vanderburg, A., Jankowski, A., et al. incl. **Yarza, R.** 2024, [AJ, 167, 54](#)
TESS Hunt for Young and Maturing Exoplanets (THYME). XI. An Earth-sized Planet Orbiting a Nearby, Solar-like Host in the 400 Myr Ursa Major Moving Group
6. **Yarza, R.**, [Razo-López, N. B.](#), Murguía-Berthier, A., Everson, R. W., et al. 2023, [ApJ, 954, 176](#)
Hydrodynamics and Survivability during Post-main-sequence Planetary Engulfment
5. Aleo, P. D., Malanchev, K., Sharief, S., Jones, D. O., et al. incl. **Yarza, R.** 2023, [ApJS, 266, 9](#)
The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae
4. Liu, C., Mockler, B., Ramirez-Ruiz, E., **Yarza, R.**, et al. 2023, [ApJ, 944, 184](#)
Tidal Disruption Events from Eccentric Orbits and Lessons Learned from the Noteworthy ASASSN-14ko
3. Wong, G. N., Prather, B. S., Dhruv, V., Ryan, B. R., et al. incl. **Yarza, R.** 2022, [ApJS, 259, 64](#)
PATOKA: Simulating Electromagnetic Observables of Black Hole Accretion
2. **Yarza, R.**, Wong, G. N., Ryan, B. R., & Gammie, C. F. 2020, [ApJ, 898, 50](#)
Bremsstrahlung in GRMHD Models of Accreting Black Holes
1. Tovmassian, G., Szkody, P., **Yarza, R.**, & Kennedy, M. 2018, [ApJ, 863, 47](#)
K2 Study of the Magnetic Precataclysmic Variable V1082 Sagittarius

Other publications (e.g, software, conference abstracts):

14. Jenkins, S., Vanderburg, A., Becker, J., Mang, J., et al. incl. **Yarza, R.** 2026, [AAS, 247, 428.04](#)
Probing the Dynamical Histories of White Dwarf Planets: Atmospheric Signatures of Common Envelope Evolution
13. Lane, K., Stephan, A., Soares-Furtado, M., Stassun, K., & **Yarza, R.** 2026, [AAS, 247, 428.07](#)
Observable Metal Pollution in Main-Sequence Stars: Simulation of Rocky Planets Engulfed by Stars in the 0.5 to 1.4 M_{sol} Range
12. Meza, A., Rosselli-Calderon, A., **Yarza, R.**, Trani, A., et al. 2026, [AAS, 247, 372.03](#)
Harnessing Stellar Chaos: Exploring the Fates of Binary-Single Encounters Embedded in Common Envelopes

11. Bush, R. C., Wu, S. C., Wallace Everson, R., **Yarza, R.**, et al. 2025, [AAS, 246, 212.07](#)
Imposters in the Galactic Center: The Unusual Early-Time Appearance of Tidal Disruption Remnants
10. Lane, K., Stephan, A., Soares-Furtado, M., Stassun, K., & **Yarza, R.** 2025, [DDA, 56, 204.12](#)
A Detailed Model of Planetary Engulfment by Main Sequence Stars: Evolution of the Planetary Orbit and Observable Stellar Signatures
9. Stephan, A., Lane, K., Soares-Furtado, M., Stassun, K., & **Yarza, R.** 2025, [DDA, 56, 100.03](#)
The Death Spiral of Doomed Worlds: Orbital Evolution of Planets Engulfed by Stars
8. Kotten, B., Soares-Furtado, M., **Yarza, R.**, Nine, A., & Ramirez-Ruiz, E. 2024, [AAS, 244, 101.05](#)
Chemical Abundance Tracers in Planetary Engulfment Candidate
7. Rohoza, V., Rosselli-Calderon, A., **Yarza, R.**, Murguia Berthier, A., & Ramirez-Ruiz, E. 2024, [HEAD, 21, 501.01](#)
Preparing Ingredients for Black Hole Mergers: Fate of Binary Black Holes Traveling through Gaseous Media at Different Inclinations
6. Caddy, R., Schneider, E., Villaseñor, B., Richie, H., et al. incl. **Yarza, R.** 2024, [Zenodo bcaddy/cholla: Caddy & Schneider 2024](#)
5. **Yarza, R.** 2024, [APS, 2024, R09.003](#)
Modeling Planetary Engulfment
4. Kotten, B., Soares-Furtado, M., **Yarza, R.**, Nine, A., & Ramirez-Ruiz, E. 2024, [AAS, 243, 205.11](#)
Chemical Abundance Tracers in Planetary Engulfment Candidate
3. **Yarza, R.** 2023, [DDA, 55, 302.01](#)
The hydrodynamics of planetary engulfment
2. Aleo, P. D., Malanchev, K., Sharief, S., Jones, D. O., et al. incl. **Yarza, R.** 2023, [yCat, 226, J/ApJS/266/9](#)
VizieR Online Data Catalog: Young Supernova Experiment (YSE) DR1 (Aleo+, 2023)
1. **Yarza, R.** 2022, [AAS, 240, 132.05](#)
The hydrodynamics of planetary engulfment